Task 1 — Augmented Lagrangian solver (plain explanation)

1.1 Problem paraphrase (simple language)

We want a program that solves optimization problems with equality and/or inequality constraints.
A typical problem form is:

Minimize the function $f(x)$ subject to constraints $c_i(x) = 0$ (equalities) and $g_j(x) \leq 0$ (inequalities).
In plain words: "find the best x that makes $f(x)$ as small as possible while satisfying the required rules."

1.2 Using Augmented Lagrangian (AL):

Augmented Lagrangian is a standard method that turns a constrained problem into a sequence of *unconstrained* problems.
Instead of solving the constraint problem directly, using AL we build a new function

$$L_\mu(x, \lambda) = f(x) + \lambda^T c(x) + \frac{\mu}{2}\|c(x)\|^2,$$

where λ is a vector of Lagrange multiplier estimates and $\mu > 0$ is a penalty weight.

Algorithm idea (brief):

1. Start with a guess $x^{(0)}$, multipliers $\lambda^{(0)}$ and penalty $\mu > 0$.
2. Repeat:
   - (Inner) Minimize $L_\mu(x, \lambda)$ with respect to x using an *unconstrained* solver (I need to call BFGS solver from project 1 or L-BFGS from scipy package).
   - Update multipliers by $\lambda \leftarrow \lambda + \mu\, c(x)$.
   - Optionally increase μ if constraint violation is still large.
   - Stop when constraints are satisfied (small $\|c(x)\|$) and x changes little.

Why it is simple to use: each inner step is an unconstrained problem — we can plug any robust unconstrained optimizer (BFGS, L-BFGS, trust-region, etc.). The AL outer loop enforces constraints.

1.2 Pseudocode (short)

Initialize x, lambda, mu

for k in 0..Kmax:

x = UnconstrainedMinimize(L_mu(x, lambda)) # inner solve

res = c(x)

if ||res|| < tol: break

lambda = lambda + mu * res

(optional) increase mu

return x, lambda

1.3 The inner solver and expected return values

We will call an unconstrained solver that has this contract:

x_opt, info = unconstrained_solver(fun, grad, x0, options)

Where:

- `fun(x)` returns scalar objective value.
- `grad(x)` returns gradient vector (same length as `x`).
- `x_opt` is the found minimizer.
- `info` is a dictionary with at least:
  - `success`: True/False
  - `nit`: number of iterations
  - `message`: short string

The provided code cell below implements:

- an Augmented Lagrangian driver `augmented_lagrangian(...)` that calls an inner solver, and
- our BFGS + Armijo line search from project 1 is used (small, easy-to-read) when `scipy` is not available. We need to call our own unconstrained optimizer by matching the expected contract.

In [ ]:
```
# Task 1: Augmented Lagrangian + inner unconstrained solver (BFGS fallback)
# This cell provides:
# - a small BFGS+Armijo unconstrained solver (used if scipy not available)
# - an augmented_lagrangian() driver that calls the inner solver
# - clear return-value structure so you can replace the inner solver with your own

import numpy as np
from math import sqrt

# -----------------------
```

```python
# Small utility: Armijo backtracking line search
# ---------- Hardened Armijo line search ----------
def armijo_line_search_safe(phi, x, p, grad, alpha0=1.0, c=1e-4, tau=0.5, min_alpha=1e-12,
    """Armijo that detects nan/inf and reduces alpha; returns alpha or 0 if failed."""
    alpha = alpha0
    try:
        fx = phi(x)
    except Exception:
        fx = np.nan
    gtp = float(np.dot(grad, p))
    for it in range(max_iter):
        x_new = x + alpha * p
        try:
            fnew = phi(x_new)
        except Exception:
            fnew = np.nan
        # if fnew is not finite, reduce alpha and continue
        if not np.isfinite(fnew):
            alpha *= tau
            if alpha < min_alpha:
                return 0.0
            continue
        # Armijo condition
        if fnew <= fx + c * alpha * gtp:
            return alpha
        alpha *= tau
        if alpha < min_alpha:
            return 0.0
    return 0.0

# -----------------------
# Simple BFGS unconstrained optimizer (returns x, info)
# ---------- Hardened BFGS fallback (with NaN/Inf checks) ----------
def bfgs_unconstrained_safe(fun, grad, x0, maxiter=300, tol=1e-8):
    """
    BFGS with extra safeguards: rejects steps that produce NaN/Inf, resets H if update inva
    Returns x_opt, info dict
    """
    n = len(x0)
    x = x0.astype(float).copy()
    I = np.eye(n)
    H = I.copy()
    fx = fun(x)
    gx = grad(x)
    success = False

    # check initial finiteness
    if not (np.isfinite(fx) and np.all(np.isfinite(gx))):
        return x, {'success': False, 'nit': 0, 'message': 'initial non-finite fun/grad'}

    for k in range(1, maxiter+1):
        gnorm = np.linalg.norm(gx)
        if gnorm < tol:
            success = True
            break
```

```
        # compute direction, ensure it's descent and finite
        p = -H.dot(gx)
        if not np.all(np.isfinite(p)) or np.dot(p, gx) >= 0:
            p = -gx  # fallback steepest descent

        # line search (safe)
        alpha = armijo_line_search_safe(lambda z: fun(z), x, p, gx, alpha0=1.0)
        if alpha == 0.0:
            # failed to find usable alpha; try tiny step
            alpha = 1e-8
            x_trial = x + alpha * (-gx)
            f_trial = fun(x_trial)
            if not np.isfinite(f_trial):
                return x, {'success': False, 'nit': k, 'message': 'line-search failed (non-
            alpha = 1e-8  # accept tiny step

        s = alpha * p
        x_new = x + s
        gx_new = grad(x_new)

        # check finiteness
        if not (np.all(np.isfinite(x_new)) and np.all(np.isfinite(gx_new)) and np.isfinite(
            # reject update and try resetting H
            H = I.copy()
            # try small steepest descent step instead
            s = 1e-8 * (-gx)
            x_new = x + s
            gx_new = grad(x_new)
            if not np.all(np.isfinite(gx_new)):
                return x, {'success': False, 'nit': k, 'message': 'non-finite gradient afte
        y = gx_new - gx
        ys = np.dot(y, s)

        # BFGS update only when ys sufficiently large and finite
        if np.isfinite(ys) and abs(ys) > 1e-12:
            # standard BFGS formula for inverse Hessian update
            try:
                V = np.outer(s, y) / ys
                H = (I - V).dot(H).dot(I - V.T) + np.outer(s, s) / ys
            except Exception:
                H = I.copy()
        else:
            # reset H to identity to maintain stability
            H = I.copy()

        x, gx = x_new, gx_new
        fx = fun(x)

    info = {'success': success, 'nit': k, 'message': 'bfgs_safe' if success else 'maxiter_o
    return x, info


# -----------------------
# Augmented Lagrangian driver
# -----------------------
def augmented_lagrangian(obj_fun, obj_grad, c_eq, c_eq_jac, x0,
```

```
                        inner_solver=bfgs_unconstrained_safe,
                        lambda0=None, mu0=1.0,
                        mu_mult=10.0,
                        max_outer=50, tol_eq=1e-8, verbose=False):
    """
    obj_fun(x) -> scalar
    obj_grad(x) -> gradient vector
    c_eq(x) -> residual vector for equalities (m,)
    c_eq_jac(x) -> Jacobian matrix of equalities (m,n) or function returning same
    inner_solver: function with signature (fun, grad, x0) -> (x_opt, info)
                  where fun and grad are callables for the inner objective.
                  If None, uses bfgs_unconstrained defined above.
    lambda0: initial multipliers (length m). If None uses zeros.
    mu0: initial penalty
    Returns: x, lambda, mu, info
    """
    if inner_solver is None:
        inner_solver = bfgs_unconstrained_safe

    m = len(c_eq(x0))
    x = x0.astype(float).copy()
    lam = np.zeros(m) if lambda0 is None else lambda0.astype(float).copy()
    mu = float(mu0)

    def L_and_grad(x_curr):
        # AL objective and gradient for inner solver
        f = obj_fun(x_curr)
        g = obj_grad(x_curr)
        ce = c_eq(x_curr)
        # Jacobian (m x n)
        J = c_eq_jac(x_curr)
        L = f + float(np.dot(lam, ce)) + 0.5 * mu * float(np.dot(ce, ce))
        gradL = g + J.T.dot(lam + mu * ce)
        return L, gradL

    info_history = []
    prev_eq_norm = np.linalg.norm(c_eq(x))
    for k in range(max_outer):
        # wrap L and grad for inner solver signature
        def fun_inner(z):
            Lval, _ = L_and_grad(z)
            return float(Lval)
        def grad_inner(z):
            _, gradL = L_and_grad(z)
            return gradL

        x_opt, inner_info = inner_solver(fun_inner, grad_inner, x.copy())
        x = x_opt.copy()
        eq_res = c_eq(x)
        eq_norm = np.linalg.norm(eq_res)
        fval = obj_fun(x)
        info_history.append({'outer_iter': k, 'x': x.copy(), 'f': fval, 'eq_norm': eq_norm,

        if verbose:
            print(f"[AL iter {k:2d}] f={fval:.6e}, ||c(x)||={eq_norm:.3e}, mu={mu:.1e}, inn
```

```python
        # convergence check on equality residuals
        if eq_norm < tol_eq:
            break

        # multiplier update
        lam = lam + mu * eq_res

        # adapt mu heuristics: increase if residual not decreasing enough
        if eq_norm > 0.5 * prev_eq_norm:
            mu = mu * mu_mult
        prev_eq_norm = eq_norm

    info = {'history': info_history, 'outer_iters': len(info_history)}
    return x, lam, mu, info

# -----------------------
# Example usage template (you will replace obj_fun, obj_grad, c_eq, c_eq_jac)
# -----------------------
if __name__ == "__main__":
    # tiny test problem: minimize (x1-1)^2 + (x2-2)^2 subject to x1 + x2 = 3
    def f_test(x): return (x[0]-1.0)**2 + (x[1]-2.0)**2
    def g_test(x): return np.array([2*(x[0]-1.0), 2*(x[1]-2.0)])
    def ce_test(x): return np.array([x[0] + x[1] - 3.0])
    def Jce_test(x): return np.array([[1.0, 1.0]])  # shape (1,2)

    x0 = np.array([0.5, 0.5])
    x_sol, lam_sol, mu_final, info = augmented_lagrangian(
        obj_fun=f_test, obj_grad=g_test,
        c_eq=ce_test, c_eq_jac=Jce_test,
        x0=x0, inner_solver=bfgs_unconstrained_safe,
        mu0=1.0, mu_mult=10.0, max_outer=30, tol_eq=1e-8, verbose=True
    )
    print("\nSolution:", x_sol, "lambda:", lam_sol, "mu:", mu_final)
    print("Info keys:", info.keys())
```

```
[AL iter  0] f=1.280485e-22, ||c(x)||=2.046e-12, mu=1.0e+00, inner_success=True

Solution: [1. 2.] lambda: [0.] mu: 1.0
Info keys: dict_keys(['history', 'outer_iters'])
```

Task 2 — Objective Function and Constraint Definitions

The chemical equilibrium problem requires minimizing the Gibbs free energy

$$f(x) = \sum_{j=1}^{10} x_j \left(c_j + \ln\left(\frac{x_j}{s}\right)\right),$$

where

$$s = \sum_{j=1}^{10} x_j.$$

Because the mixture contains strictly positive mole quantities, we enforce a numerically safe objective by replacing the undefined terms $\ln(0)$ and $1/s$ with smooth lower-bounds:

- Replace all x_j with $\max(x_j, \varepsilon)$
- Replace s with $\max(s, \varepsilon)$
- Use a small constant $\varepsilon = 10^{-12}$

Thus the implemented objective is

$$\tilde{f}(x) = \sum_{j=1}^{10} x_j \left(c_j + \ln\left(\frac{\max(x_j, \varepsilon)}{\max\left(\sum_{k=1}^{10} x_k,\ \varepsilon\right)} \right) \right).$$

This preserves the true Gibbs energy for all feasible physical points but remains well-defined during the intermediate iterations of the optimization process (where the AL method must evaluate possibly infeasible points).

Equality Constraints

The chemical mixture must respect atomic conservation for Hydrogen (H), Nitrogen (N), and Oxygen (O).
These are linear equality constraints of the form $c(x) = 0$:

Hydrogen balance

$$x_1 + 2x_2 + 2x_3 + x_6 + x_{10} = 4$$

Nitrogen balance

$$x_4 + 2x_5 + x_6 + x_7 = 2$$

Oxygen balance

$$x_3 + x_7 + x_8 + 2x_9 + x_{10} = 2$$

We express these constraints in vector form:

$$c(x) = \begin{bmatrix} x_1 + 2x_2 + 2x_3 + x_6 + x_{10} - 4 \\ x_4 + 2x_5 + x_6 + x_7 - 2 \\ x_3 + x_7 + x_8 + 2x_9 + x_{10} - 2 \end{bmatrix}.$$

Jacobian of the Constraints

The Jacobian matrix $J = \partial c / \partial x$ is constant because all constraints are linear. Explicitly:

$$J = \begin{bmatrix} 1 & 2 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 2 & 1 \end{bmatrix}.$$

We will pass this vector-valued constraint function and its Jacobian to the Augmented Lagrangian solver.

Summary

- Objective uses the clipped formulation to avoid evaluating $\ln(0)$.
- Equality constraints encode exact atomic conservation.
- Constraint Jacobian is constant, simplifying implementation.
- Works with the given inner solver or AL + BFGS (unconstrained) or projected gradient decent as a fall back.

In [1]:

```
# (AL outer + L-BFGS-B inner with clipped-log objective).
# Focus: use the H2O/N2 hint initial guess, stable gradients, friendly prints sorted by mol
import numpy as np

# -------------------------
# Problem data (self-contained)
# -------------------------
names = ["H","H2","H2O","N","N2","NH","NO","O","O2","OH"]
c = np.array([-6.089, -17.164, -34.054, -5.914, -24.721,
              -14.986, -24.100, -10.708, -26.662, -22.179], dtype=float)
n = len(c)

A = np.zeros((3, n))
A[0, 0] = 1; A[0, 1] = 2; A[0, 2] = 2; A[0, 5] = 1; A[0, 9] = 1  # H
A[1, 3] = 1; A[1, 4] = 2; A[1, 5] = 1; A[1, 6] = 1               # N
A[2, 2] = 1; A[2, 6] = 1; A[2, 7] = 1; A[2, 8] = 2; A[2, 9] = 1  # O
b = np.array([4.0, 2.0, 2.0], dtype=float)

# -------------------------
# Clipped-log objective & gradient (x-space)
# Uses tiny eps to avoid log(0). Analytical gradient (with clipped x_safe).
# -------------------------
def f_x_clipped(x, eps=1e-12):
    x = np.asarray(x, dtype=float)
    x_safe = np.maximum(x, eps)
    s = np.sum(x_safe)
    if s <= 0 or not np.isfinite(s):
```

```
        return np.inf
    ln = np.log(x_safe / s)
    return float(np.sum(x_safe * (c + ln)))

def grad_f_x_clipped(x, eps=1e-12):
    x = np.asarray(x, dtype=float)
    x_safe = np.maximum(x, eps)
    s = np.sum(x_safe)
    if s <= 0 or not np.isfinite(s):
        return np.full_like(x, np.nan)
    # gradient derived from f(x) = sum x_j (c_j + ln(x_j/s))
    # df/dx_j = c_j + ln(x_j/s) + 1 - x_j/s (use x_safe where needed)
    return c + np.log(x_safe / s) + 1.0 - x_safe / s

# -------------------------
# Linear equality constraints
# -------------------------
def c_eq(x):
    return A.dot(x) - b

def c_eq_jac(x):
    return A

# -------------------------
# Inner solver: prefer scipy L-BFGS-B; otherwise use PGD fallback
# -------------------------
use_scipy = True
try:
    from scipy.optimize import minimize
except Exception:
    use_scipy = False

def inner_solver_lbfgsb(fun, jac, x0, eps=1e-12, options=None):
    bounds = [(eps, None)] * len(x0)
    res = minimize(fun, x0, method='L-BFGS-B', jac=jac, bounds=bounds,
                   options=(options or {'ftol':1e-12, 'maxiter':1000}))
    return res.x, {'success': res.success, 'nit': getattr(res, 'nit', None), 'message': res

def projected_gradient_descent(fun, grad, x0, eps=1e-12, maxiter=20000, alpha0=1e-2, tol=1e
    x = np.maximum(x0.astype(float), eps)
    fval = fun(x)
    for k in range(1, maxiter+1):
        g = grad(x)
        gnorm = np.linalg.norm(g)
        if gnorm < tol:
            return x, {'success': True, 'nit': k, 'message': 'pgd_converged'}
        alpha = alpha0
        while alpha > 1e-16:
            x_new = np.maximum(x - alpha * g, eps)
            f_new = fun(x_new)
            if np.isfinite(f_new) and f_new <= fval - 1e-4 * alpha * (gnorm**2):
                break
            alpha *= 0.5
        if alpha <= 1e-16:
            return x, {'success': False, 'nit': k, 'message': 'pgd_linefail'}
        x, fval = x_new, f_new
```

```
    return x, {'success': False, 'nit': maxiter, 'message': 'pgd_maxiter'}

def inner_solver(fun, jac, x0, eps=1e-12):
    if use_scipy:
        return inner_solver_lbfgsb(fun, jac, x0, eps=eps)
    else:
        return projected_gradient_descent(fun, jac, x0, eps=eps)

# -------------------------
# Minimal robust Augmented Lagrangian (x-space)
# - simple mu schedule, numeric safeguards, prints
# -------------------------
def augmented_lagrangian_x(obj_fun, obj_grad, c_eq, c_eq_jac, x0,
                           inner_solver_func=inner_solver,
                           lambda0=None, mu0=10.0, mu_mult=10.0,
                           max_outer=60, tol_eq=1e-9, eps=1e-12, verbose=False):
    m = len(c_eq(x0))
    x = np.maximum(x0.astype(float), eps)
    lam = np.zeros(m) if lambda0 is None else lambda0.astype(float).copy()
    mu = float(mu0)
    history = []
    prev_eq_norm = np.linalg.norm(c_eq(x))
    for k in range(max_outer):
        def L_and_grad(xx):
            f = obj_fun(xx, eps=eps)
            g = obj_grad(xx, eps=eps)
            ce = c_eq(xx)
            J = c_eq_jac(xx)
            if not (np.isfinite(f) and np.all(np.isfinite(g)) and np.all(np.isfinite(ce))):
                return np.inf, np.full_like(xx, np.inf)
            L = f + float(np.dot(lam, ce)) + 0.5 * mu * float(np.dot(ce, ce))
            gradL = g + J.T.dot(lam + mu * ce)
            return L, gradL

        fun_inner = lambda xx: float(L_and_grad(xx)[0])
        grad_inner = lambda xx: L_and_grad(xx)[1]

        x_opt, inner_info = inner_solver_func(fun_inner, grad_inner, x.copy(), eps=eps)
        x = np.maximum(x_opt, eps)  # enforce lower bound numerically
        eq_res = c_eq(x)
        eq_norm = np.linalg.norm(eq_res)
        fval = obj_fun(x, eps=eps)
        history.append({'outer_iter': k, 'x': x.copy(), 'f': fval, 'eq_norm': eq_norm, 'mu'

        if verbose:
            print(f"[AL {k:2d}] f={fval:.6e}, ||Ax-b||={eq_norm:.3e}, mu={mu:.1e}, inner_su

        if eq_norm < tol_eq:
            break

        lam = lam + mu * eq_res

        # modest mu increase only when progress stalls or inner fails
        if not inner_info.get('success', True) or eq_norm > 0.7 * prev_eq_norm:
            mu = min(mu * mu_mult, 1e12)
        prev_eq_norm = eq_norm
```

```python
    return x, lam, mu, history

# -------------------------
# Clean run using the H2O & N2 hint (single focused run)
# -------------------------
def clean_run(eps=1e-12, verbose=True):
    # Hint initial guess: H2O (index 2) and N2 (index 4) larger, others small positive
    x0 = np.ones(n) * 1e-6
    # x0[2] = 1.6   # H2O (component 3)
    # x0[4] = 0.98  # N2  (component 5)
    # ensure feasibility approx (we don't enforce exact eq initially)
    x0 = np.maximum(x0, eps)

    x_sol, lam, mu_final, history = None, None, None, None
    try:
        x_sol, lam, mu_final, history = None, None, None, None
        x_sol, lam, mu_final, history = augmented_lagrangian_x(
            obj_fun=f_x_clipped, obj_grad=grad_f_x_clipped,
            c_eq=c_eq, c_eq_jac=c_eq_jac,
            x0=x0, inner_solver_func=inner_solver,
            lambda0=None, mu0=10.0, mu_mult=10.0,
            max_outer=80, tol_eq=1e-9, eps=eps, verbose=verbose
        )
    except Exception as e:
        print("AL run failed with exception:", e)
        return None

    # compute original objective (with tiny safeguard)
    fval = f_x_clipped(x_sol, eps=eps)
    eq_res = c_eq(x_sol)
    # sort species by amount descending
    order = np.argsort(-x_sol)
    if verbose:
        print("\n=== Clean Run Results ===")
        print(f"Objective f = {fval:.6e}")
        print(f"Sum of moles s = {np.sum(x_sol):.6e}")
        print("Equality residuals A x - b =", eq_res)
        print("\nTop species (index, name, mole):")
        for i in order[:8]:
            print(f"  {i+1:2d} {names[i]:>4s}  {x_sol[i]:.6e}")
    return {'x': x_sol, 'f': fval, 'eq_res': eq_res, 'order': order, 'lam': lam, 'mu_final'

# -------------------------
# Execute clean run
# -------------------------
res = clean_run(eps=1e-12, verbose=True)

# If you want to inspect full vector:
if res is not None:
    print("\nFull solution vector x_j (index, name, value):")
    for i in range(n):
        print(f"  x{i+1:2d} {names[i]:>4s} = {res['x'][i]:.8e}")
```

```
[AL  0] f=-1.433169e+02, ||Ax-b||=2.166e+00, mu=1.0e+01, inner_success=True
[AL  1] f=-9.567543e+01, ||Ax-b||=8.852e-02, mu=1.0e+01, inner_success=False
[AL  2] f=-9.558583e+01, ||Ax-b||=5.598e-03, mu=1.0e+02, inner_success=True
[AL  3] f=-9.545553e+01, ||Ax-b||=5.610e-03, mu=1.0e+02, inner_success=False
[AL  4] f=-9.550606e+01, ||Ax-b||=3.847e-04, mu=1.0e+03, inner_success=True
[AL  5] f=-9.550149e+01, ||Ax-b||=2.255e-04, mu=1.0e+03, inner_success=True
[AL  6] f=-9.549890e+01, ||Ax-b||=1.572e-04, mu=1.0e+03, inner_success=True
[AL  7] f=-9.550111e+01, ||Ax-b||=4.282e-04, mu=1.0e+03, inner_success=True
[AL  8] f=-9.550274e+01, ||Ax-b||=4.947e-05, mu=1.0e+04, inner_success=True
[AL  9] f=-9.550139e+01, ||Ax-b||=3.996e-05, mu=1.0e+04, inner_success=True
[AL 10] f=-9.550215e+01, ||Ax-b||=4.000e-06, mu=1.0e+05, inner_success=True
[AL 11] f=-9.550224e+01, ||Ax-b||=2.861e-06, mu=1.0e+05, inner_success=True
[AL 12] f=-9.550218e+01, ||Ax-b||=1.927e-07, mu=1.0e+06, inner_success=True
[AL 13] f=-9.550219e+01, ||Ax-b||=2.423e-07, mu=1.0e+06, inner_success=True
[AL 14] f=-9.550218e+01, ||Ax-b||=5.912e-08, mu=1.0e+07, inner_success=True
[AL 15] f=-9.550218e+01, ||Ax-b||=3.786e-08, mu=1.0e+07, inner_success=True
[AL 16] f=-9.550218e+01, ||Ax-b||=2.758e-08, mu=1.0e+07, inner_success=True
[AL 17] f=-9.550218e+01, ||Ax-b||=5.915e-09, mu=1.0e+08, inner_success=True
[AL 18] f=-9.550218e+01, ||Ax-b||=4.476e-09, mu=1.0e+08, inner_success=True
[AL 19] f=-9.550218e+01, ||Ax-b||=1.847e-10, mu=1.0e+09, inner_success=True

=== Clean Run Results ===
Objective f = -9.550218e+01
Sum of moles s = 3.222220e+00
Equality residuals A x - b = [-1.57104996e-10 -6.50428600e-11 -7.20190574e-11]

Top species (index, name, mole):
   3  H2O  1.636856e+00
   5   N2  9.762132e-01
   2   H2  2.366619e-01
  10   OH  1.945379e-01
   1    H  5.659454e-02
   9   O2  5.123123e-02
   7   NO  4.359232e-02
   8    O  2.255087e-02

Full solution vector x_j (index, name, value):
  x 1    H = 5.65945372e-02
  x 2   H2 = 2.36661925e-01
  x 3  H2O = 1.63685647e+00
  x 4    N = 2.15042707e-03
  x 5   N2 = 9.76213235e-01
  x 6   NH = 1.83078529e-03
  x 7   NO = 4.35923178e-02
  x 8    O = 2.25508655e-02
  x 9   O2 = 5.12312287e-02
  x10   OH = 1.94537890e-01
```

Task 3 — Solve & Sensitivity Study (instructions)

Goal: solve the chemical equilibrium problem with the Augmented Lagrangian (AL) code you have and explore how the final solution depends on (a) initial guesses and (b)

hyperparameters (penalty `mu0`, multiplier `mu_mult`, inner-solver tolerances). Report objective values, dominant species, and sensitivity observations.

---

1) Problem summary (reminder)

We minimize the Gibbs free energy

$$f(x) = \sum_{j=1}^{10} x_j \left(c_j + \ln \frac{x_j}{s}\right), \qquad s = \sum_{j=1}^{10} x_j,$$

subject to linear atomic balances

$$Ax = b,$$

with positivity enforced numerically as $x_j \geq \varepsilon$.

We use the clipped-log objective (replace x_j by $\max(x_j, \varepsilon)$ inside $\ln$) and the AL outer loop with a bound-aware inner solver (L-BFGS-B or projected gradient descent fallback).

In [2]:
```
# ===== Minimal safe patch: coerce inputs/outputs to 1-D float arrays =====
import numpy as np

# Defensive wrappers for constraints (coerce x to 1-D numeric array)
def c_eq_safe(x):
    x = np.asarray(x, dtype=float).ravel()
    return A.dot(x) - b

def c_eq_jac_safe(x):
    _ = np.asarray(x, dtype=float).ravel()
    return A

# Defensive augmented Lagrangian wrapper that coerces x0 and x_opt
def augmented_lagrangian_x_safe(obj_fun, obj_grad, c_eq_func, c_eq_jac_func, x0,
                                inner_solver_func=inner_solver,
                                lambda0=None, mu0=10.0, mu_mult=10.0,
                                max_outer=60, tol_eq=1e-9, eps=1e-12, verbose=False):
    # coerce x0
    x0 = np.asarray(x0, dtype=float).ravel()
    # call original function but pass safe c_eq/c_eq_jac
    x_sol, lam, mu, history = augmented_lagrangian_x(
        obj_fun=obj_fun, obj_grad=obj_grad,
        c_eq=c_eq_func, c_eq_jac=c_eq_jac_func,
        x0=x0, inner_solver_func=inner_solver_func,
        lambda0=lambda0, mu0=mu0, mu_mult=mu_mult,
        max_outer=max_outer, tol_eq=tol_eq, eps=eps, verbose=verbose
    )
    # coerce outputs
    x_sol = np.asarray(x_sol, dtype=float).ravel()
    lam = np.asarray(lam, dtype=float).ravel()
    return x_sol, lam, mu, history
```

```python
def run_experiments(eps=1e-12, verbose_al=True): # verbose_al changed to True as requested
    # Hint initial guess: H2O (index 2) and N2 (index 4) larger, others small positive
    initial_guesses = {}
    initial_guesses['uniform_1'] = np.ones(n)
    initial_guesses['uniform_0.5'] = np.ones(n)/2
    initial_guesses['uniform_3'] = np.ones(n)*3
    # Adding the hinted guess from previous runs
    # x_hint = np.ones(n) * 1e-6
    # x_hint[2] = 1.6 # H2O (index 2)
    # x_hint[4] = 0.98 # N2 (index 4)
    # initial_guesses['hint_H2O_N2'] = x_hint

    results_x0s = {}

    mu_mults = [5, 10, 100]
    results_mu_mults = {}

    al_eps_values = [1e-6, 1e-8, 1e-10, 1e-14]
    results_al_eps = {}

    # Getting results for different x0s
    print("\n--- Running experiments for different initial guesses (x0) ---")
    for name, x0 in initial_guesses.items():
        print(f"  Running with x0: {name}")
        try:
            x_sol, lam, mu_final, history = augmented_lagrangian_x_safe(
                obj_fun=f_x_clipped, obj_grad=grad_f_x_clipped,
                c_eq_func=c_eq_safe, c_eq_jac_func=c_eq_jac_safe,
                x0=x0, inner_solver_func=inner_solver,
                lambda0=None, mu0=10.0, mu_mult=10.0,
                max_outer=80, tol_eq=1e-9, eps=eps, verbose=verbose_al
            )
            # compute original objective (with tiny safeguard)
            fval = f_x_clipped(x_sol, eps=eps)
            eq_res = c_eq_safe(x_sol)
            results_x0s[name] = {'x0_name': name, 'x': x_sol, 'f': fval, 'eq_res': eq_res,
        except Exception as e:
            print(f"AL run failed for x0 '{name}' with exception: {e}")
            results_x0s[name] = None

    # Setting a default x0 for the rest of the program (e.g., 'hint_H2O_N2' if it exists)
    default_x0_name = 'hint_H2O_N2' if 'hint_H2O_N2' in initial_guesses else 'uniform_1'
    x0_default = initial_guesses[default_x0_name]

    # Getting results for different mu_mults
    print(f"\n--- Running experiments for different mu_mults (using x0=\'{default_x0_name}\
    for mu_mult_val in mu_mults:
        print(f"  Running with mu_mult: {mu_mult_val}")
        try:
            x_sol, lam, mu_final, history = augmented_lagrangian_x_safe(
                obj_fun=f_x_clipped, obj_grad=grad_f_x_clipped,
                c_eq_func=c_eq_safe, c_eq_jac_func=c_eq_jac_safe,
                x0=x0_default, inner_solver_func=inner_solver,
                lambda0=None, mu0=10.0, mu_mult=float(mu_mult_val),
                max_outer=80, tol_eq=1e-9, eps=eps, verbose=verbose_al
```

```
            )
            # compute original objective (with tiny safeguard)
            fval = f_x_clipped(x_sol, eps=eps)
            eq_res = c_eq_safe(x_sol)
            results_mu_mults[mu_mult_val] = {'mu_mult': mu_mult_val, 'x': x_sol, 'f': fval,
        except Exception as e:
            print(f"AL run failed for mu_mult '{mu_mult_val}' with exception: {e}")
            results_mu_mults[mu_mult_val] = None

    # Getting results for different al_eps's
    print(f"\n--- Running experiments for different clipping epsilons (using x0=\'{default_
    for current_eps in al_eps_values:
        print(f"  Running with eps: {current_eps}")
        try:
            x_sol, lam, mu_final, history = augmented_lagrangian_x_safe(
                obj_fun=f_x_clipped, obj_grad=grad_f_x_clipped,
                c_eq_func=c_eq_safe, c_eq_jac_func=c_eq_jac_safe,
                x0=x0_default, inner_solver_func=inner_solver,
                lambda0=None, mu0=10.0, mu_mult=10.0,
                max_outer=80, tol_eq=1e-9, eps=current_eps, verbose=verbose_al
            )
            # compute original objective (with tiny safeguard)
            fval = f_x_clipped(x_sol, eps=current_eps)
            eq_res = c_eq_safe(x_sol)
            results_al_eps[current_eps] = {'eps': current_eps, 'x': x_sol, 'f': fval, 'eq_r
        except Exception as e:
            print(f"AL run failed for eps '{current_eps}' with exception: {e}")
            results_al_eps[current_eps] = None

    all_results = {'x0_experiments': results_x0s,
                   'mu_mult_experiments': results_mu_mults,
                   'epsilon_experiments': results_al_eps}
    return all_results

# -------------------------
# Execute run experiments and store results globally
# -------------------------
full_results = run_experiments(eps=1e-12, verbose_al=True)

# Print a summary of all results
print("\n=== Summary of All Experiments ===")

print("\n--- Initial Guess (x0) Experiments ---")
for name, res in full_results['x0_experiments'].items():
    if res:
        print(f"  x0: {name}, Objective f: {res['f']:.6e}, ||Ax-b||: {np.linalg.norm(res['e
    else:
        print(f"  x0: {name}, Run failed")

print("\n--- Mu Multiplier (mu_mult) Experiments ---")
for mu_mult, res in full_results['mu_mult_experiments'].items():
    if res:
        print(f"  mu_mult: {mu_mult}, Objective f: {res['f']:.6e}, ||Ax-b||: {np.linalg.nor
    else:
        print(f"  mu_mult: {mu_mult}, Run failed")
```

```
print("\n--- Epsilon (eps) Experiments ---")
for eps_val, res in full_results['epsilon_experiments'].items():
    if res:
        print(f"  eps: {eps_val:.1e}, Objective f: {res['f']:.6e}, ||Ax-b||: {np.linalg.nor
    else:
        print(f"  eps: {eps_val:.1e}, Run failed")

# Example of how to inspect a specific result:
print("\n=== Detailed Inspection of a Specific Result (e.g., 'uniform_1' x0) ===")
if 'uniform_1' in full_results['x0_experiments'] and full_results['x0_experiments']['unifor
    detailed_res = full_results['x0_experiments']['uniform_1']
    print(f"Objective f = {detailed_res['f']:.6e}")
    print(f"Equality residuals A x - b = {detailed_res['eq_res']}")
    print("Top species (index, name, mole):")
    # Assuming 'n' and 'names' are defined globally from previous cells
    if 'n' in globals() and 'names' in globals():
        order = np.argsort(-detailed_res['x'])
        for i in order[:8]: # Print top 8 species
            print(f"  {i+1:2d} {names[i]:>4s}  {detailed_res['x'][i]:.6e}")
    else:
        print("  (Cannot display species details, 'n' or 'names' not defined)")
else:
    print("  Result for 'uniform_1' not available or run failed.")

print("\n=== Detailed Inspection of another result (e.g., mu_mult=100) ===")
if 100 in full_results['mu_mult_experiments'] and full_results['mu_mult_experiments'][100]:
    detailed_res_mu = full_results['mu_mult_experiments'][100]
    print(f"Objective f = {detailed_res_mu['f']:.6e}")
    print(f"Equality residuals A x - b = {detailed_res_mu['eq_res']}")
    # Assuming 'n' and 'names' are defined globally from previous cells
    if 'n' in globals() and 'names' in globals():
        order = np.argsort(-detailed_res_mu['x'])
        print("Top species (index, name, mole):")
        for i in order[:8]: # Print top 8 species
            print(f"  {i+1:2d} {names[i]:>4s}  {detailed_res_mu['x'][i]:.6e}")
    else:
        print("  (Cannot display species details, 'n' or 'names' not defined)")
else:
    print("  Result for mu_mult=100 not available or run failed.")
```

```
--- Running experiments for different initial guesses (x0) ---
  Running with x0: uniform_1
[AL  0] f=-1.428387e+02, ||Ax-b||=2.144e+00, mu=1.0e+01, inner_success=True
[AL  1] f=-9.689052e+01, ||Ax-b||=8.453e-02, mu=1.0e+01, inner_success=False
[AL  2] f=-9.550093e+01, ||Ax-b||=4.938e-03, mu=1.0e+02, inner_success=True
[AL  3] f=-9.555071e+01, ||Ax-b||=3.963e-03, mu=1.0e+02, inner_success=True
[AL  4] f=-9.550296e+01, ||Ax-b||=3.915e-04, mu=1.0e+03, inner_success=False
[AL  5] f=-9.550656e+01, ||Ax-b||=4.749e-05, mu=1.0e+04, inner_success=True
[AL  6] f=-9.550772e+01, ||Ax-b||=1.737e-05, mu=1.0e+04, inner_success=True
[AL  7] f=-9.550772e+01, ||Ax-b||=1.737e-05, mu=1.0e+04, inner_success=True
[AL  8] f=-9.550735e+01, ||Ax-b||=3.168e-06, mu=1.0e+05, inner_success=True
[AL  9] f=-9.550740e+01, ||Ax-b||=2.157e-06, mu=1.0e+05, inner_success=True
[AL 10] f=-9.550740e+01, ||Ax-b||=2.157e-06, mu=1.0e+05, inner_success=False
[AL 11] f=-9.550739e+01, ||Ax-b||=5.294e-07, mu=1.0e+06, inner_success=True
[AL 12] f=-9.550737e+01, ||Ax-b||=5.550e-07, mu=1.0e+06, inner_success=True
[AL 13] f=-9.550738e+01, ||Ax-b||=4.879e-08, mu=1.0e+07, inner_success=True
[AL 14] f=-9.550738e+01, ||Ax-b||=3.573e-08, mu=1.0e+07, inner_success=True
[AL 15] f=-9.550738e+01, ||Ax-b||=2.789e-09, mu=1.0e+08, inner_success=True
[AL 16] f=-9.550738e+01, ||Ax-b||=3.007e-09, mu=1.0e+08, inner_success=True
[AL 17] f=-9.550738e+01, ||Ax-b||=3.995e-10, mu=1.0e+09, inner_success=True
  Running with x0: uniform_0.5
[AL  0] f=-1.428615e+02, ||Ax-b||=2.147e+00, mu=1.0e+01, inner_success=False
[AL  1] f=-9.562818e+01, ||Ax-b||=1.223e-02, mu=1.0e+02, inner_success=True
[AL  2] f=-9.545274e+01, ||Ax-b||=3.172e-03, mu=1.0e+02, inner_success=True
[AL  3] f=-9.546873e+01, ||Ax-b||=2.393e-03, mu=1.0e+02, inner_success=True
[AL  4] f=-9.549485e+01, ||Ax-b||=1.813e-04, mu=1.0e+03, inner_success=True
[AL  5] f=-9.549331e+01, ||Ax-b||=7.349e-05, mu=1.0e+03, inner_success=True
[AL  6] f=-9.549331e+01, ||Ax-b||=7.349e-05, mu=1.0e+03, inner_success=True
[AL  7] f=-9.549199e+01, ||Ax-b||=3.063e-05, mu=1.0e+04, inner_success=False
[AL  8] f=-9.549173e+01, ||Ax-b||=5.525e-06, mu=1.0e+05, inner_success=True
[AL  9] f=-9.549186e+01, ||Ax-b||=2.730e-06, mu=1.0e+05, inner_success=True
[AL 10] f=-9.549183e+01, ||Ax-b||=1.731e-06, mu=1.0e+05, inner_success=True
[AL 11] f=-9.549188e+01, ||Ax-b||=1.902e-06, mu=1.0e+05, inner_success=True
[AL 12] f=-9.549185e+01, ||Ax-b||=1.931e-07, mu=1.0e+06, inner_success=True
[AL 13] f=-9.549185e+01, ||Ax-b||=1.931e-07, mu=1.0e+06, inner_success=True
[AL 14] f=-9.549184e+01, ||Ax-b||=4.270e-08, mu=1.0e+07, inner_success=True
[AL 15] f=-9.549184e+01, ||Ax-b||=2.715e-08, mu=1.0e+07, inner_success=True
[AL 16] f=-9.549184e+01, ||Ax-b||=2.715e-08, mu=1.0e+07, inner_success=False
[AL 17] f=-9.549184e+01, ||Ax-b||=5.046e-09, mu=1.0e+08, inner_success=True
[AL 18] f=-9.549184e+01, ||Ax-b||=6.093e-09, mu=1.0e+08, inner_success=True
[AL 19] f=-9.549184e+01, ||Ax-b||=2.381e-10, mu=1.0e+09, inner_success=True
  Running with x0: uniform_3
[AL  0] f=-1.440388e+02, ||Ax-b||=2.200e+00, mu=1.0e+01, inner_success=False
[AL  1] f=-9.543256e+01, ||Ax-b||=6.669e-03, mu=1.0e+02, inner_success=True
[AL  2] f=-9.554317e+01, ||Ax-b||=3.114e-03, mu=1.0e+02, inner_success=True
[AL  3] f=-9.554317e+01, ||Ax-b||=3.114e-03, mu=1.0e+02, inner_success=False
[AL  4] f=-9.550240e+01, ||Ax-b||=4.950e-04, mu=1.0e+03, inner_success=True
[AL  5] f=-9.551354e+01, ||Ax-b||=1.797e-04, mu=1.0e+03, inner_success=True
[AL  6] f=-9.551354e+01, ||Ax-b||=1.797e-04, mu=1.0e+03, inner_success=False
[AL  7] f=-9.550959e+01, ||Ax-b||=2.925e-05, mu=1.0e+04, inner_success=True
[AL  8] f=-9.550994e+01, ||Ax-b||=2.756e-05, mu=1.0e+04, inner_success=True
[AL  9] f=-9.550986e+01, ||Ax-b||=3.082e-06, mu=1.0e+05, inner_success=True
[AL 10] f=-9.550980e+01, ||Ax-b||=5.486e-07, mu=1.0e+05, inner_success=True
[AL 11] f=-9.550980e+01, ||Ax-b||=5.486e-07, mu=1.0e+05, inner_success=False
[AL 12] f=-9.550980e+01, ||Ax-b||=3.282e-07, mu=1.0e+06, inner_success=True
[AL 13] f=-9.550981e+01, ||Ax-b||=3.253e-07, mu=1.0e+06, inner_success=True
```

```
[AL 14] f=-9.550981e+01, ||Ax-b||=3.816e-08, mu=1.0e+07, inner_success=False
[AL 15] f=-9.550981e+01, ||Ax-b||=1.948e-09, mu=1.0e+08, inner_success=True
[AL 16] f=-9.550981e+01, ||Ax-b||=2.775e-09, mu=1.0e+08, inner_success=True
[AL 17] f=-9.550981e+01, ||Ax-b||=5.202e-10, mu=1.0e+09, inner_success=True

--- Running experiments for different mu_mults (using x0='uniform_1') ---
  Running with mu_mult: 5
[AL  0] f=-1.428387e+02, ||Ax-b||=2.144e+00, mu=1.0e+01, inner_success=True
[AL  1] f=-9.689052e+01, ||Ax-b||=8.453e-02, mu=1.0e+01, inner_success=False
[AL  2] f=-9.546937e+01, ||Ax-b||=4.539e-03, mu=5.0e+01, inner_success=False
[AL  3] f=-9.551204e+01, ||Ax-b||=1.300e-03, mu=2.5e+02, inner_success=False
[AL  4] f=-9.550583e+01, ||Ax-b||=1.034e-04, mu=1.2e+03, inner_success=True
[AL  5] f=-9.550852e+01, ||Ax-b||=2.723e-04, mu=1.2e+03, inner_success=True
[AL  6] f=-9.550782e+01, ||Ax-b||=8.885e-05, mu=6.2e+03, inner_success=True
[AL  7] f=-9.550472e+01, ||Ax-b||=9.967e-05, mu=6.2e+03, inner_success=True
[AL  8] f=-9.550694e+01, ||Ax-b||=4.321e-05, mu=3.1e+04, inner_success=True
[AL  9] f=-9.550698e+01, ||Ax-b||=2.974e-05, mu=3.1e+04, inner_success=True
[AL 10] f=-9.550669e+01, ||Ax-b||=1.619e-05, mu=3.1e+04, inner_success=True
[AL 11] f=-9.550696e+01, ||Ax-b||=5.351e-06, mu=3.1e+04, inner_success=True
[AL 12] f=-9.550696e+01, ||Ax-b||=5.351e-06, mu=3.1e+04, inner_success=False
[AL 13] f=-9.550683e+01, ||Ax-b||=2.116e-06, mu=1.6e+05, inner_success=True
[AL 14] f=-9.550689e+01, ||Ax-b||=1.766e-06, mu=1.6e+05, inner_success=True
[AL 15] f=-9.550687e+01, ||Ax-b||=1.365e-07, mu=7.8e+05, inner_success=True
[AL 16] f=-9.550695e+01, ||Ax-b||=1.472e-05, mu=7.8e+05, inner_success=True
[AL 17] f=-9.550696e+01, ||Ax-b||=3.009e-06, mu=3.9e+06, inner_success=True
[AL 18] f=-9.550696e+01, ||Ax-b||=1.098e-07, mu=3.9e+06, inner_success=True
[AL 19] f=-9.550696e+01, ||Ax-b||=3.780e-08, mu=3.9e+06, inner_success=True
[AL 20] f=-9.550696e+01, ||Ax-b||=3.780e-08, mu=3.9e+06, inner_success=False
[AL 21] f=-9.550696e+01, ||Ax-b||=7.791e-09, mu=2.0e+07, inner_success=True
[AL 22] f=-9.550696e+01, ||Ax-b||=1.788e-08, mu=2.0e+07, inner_success=True
[AL 23] f=-9.550696e+01, ||Ax-b||=5.187e-09, mu=9.8e+07, inner_success=True
[AL 24] f=-9.550696e+01, ||Ax-b||=5.717e-09, mu=9.8e+07, inner_success=True
[AL 25] f=-9.550696e+01, ||Ax-b||=5.511e-10, mu=4.9e+08, inner_success=True
  Running with mu_mult: 10
[AL  0] f=-1.428387e+02, ||Ax-b||=2.144e+00, mu=1.0e+01, inner_success=True
[AL  1] f=-9.689052e+01, ||Ax-b||=8.453e-02, mu=1.0e+01, inner_success=False
[AL  2] f=-9.550093e+01, ||Ax-b||=4.938e-03, mu=1.0e+02, inner_success=True
[AL  3] f=-9.555071e+01, ||Ax-b||=3.963e-03, mu=1.0e+02, inner_success=True
[AL  4] f=-9.550296e+01, ||Ax-b||=3.915e-04, mu=1.0e+03, inner_success=False
[AL  5] f=-9.550656e+01, ||Ax-b||=4.749e-05, mu=1.0e+04, inner_success=True
[AL  6] f=-9.550772e+01, ||Ax-b||=1.737e-05, mu=1.0e+04, inner_success=True
[AL  7] f=-9.550772e+01, ||Ax-b||=1.737e-05, mu=1.0e+04, inner_success=True
[AL  8] f=-9.550735e+01, ||Ax-b||=3.168e-06, mu=1.0e+05, inner_success=True
[AL  9] f=-9.550740e+01, ||Ax-b||=2.157e-06, mu=1.0e+05, inner_success=True
[AL 10] f=-9.550740e+01, ||Ax-b||=2.157e-06, mu=1.0e+05, inner_success=False
[AL 11] f=-9.550739e+01, ||Ax-b||=5.294e-07, mu=1.0e+06, inner_success=True
[AL 12] f=-9.550737e+01, ||Ax-b||=5.550e-07, mu=1.0e+06, inner_success=True
[AL 13] f=-9.550738e+01, ||Ax-b||=4.879e-08, mu=1.0e+07, inner_success=True
[AL 14] f=-9.550738e+01, ||Ax-b||=3.573e-08, mu=1.0e+07, inner_success=True
[AL 15] f=-9.550738e+01, ||Ax-b||=2.789e-09, mu=1.0e+08, inner_success=True
[AL 16] f=-9.550738e+01, ||Ax-b||=3.007e-09, mu=1.0e+08, inner_success=True
[AL 17] f=-9.550738e+01, ||Ax-b||=3.995e-10, mu=1.0e+09, inner_success=True
  Running with mu_mult: 100
[AL  0] f=-1.428387e+02, ||Ax-b||=2.144e+00, mu=1.0e+01, inner_success=True
[AL  1] f=-9.689052e+01, ||Ax-b||=8.453e-02, mu=1.0e+01, inner_success=False
[AL  2] f=-9.549035e+01, ||Ax-b||=5.759e-04, mu=1.0e+03, inner_success=True
```

```
[AL  3] f=-9.550047e+01, ||Ax-b||=3.333e-04, mu=1.0e+03, inner_success=True
[AL  4] f=-9.550659e+01, ||Ax-b||=3.422e-04, mu=1.0e+03, inner_success=False
[AL  5] f=-9.550073e+01, ||Ax-b||=3.517e-06, mu=1.0e+05, inner_success=True
[AL  6] f=-9.550086e+01, ||Ax-b||=3.986e-06, mu=1.0e+05, inner_success=True
[AL  7] f=-9.550080e+01, ||Ax-b||=4.537e-08, mu=1.0e+07, inner_success=True
[AL  8] f=-9.550080e+01, ||Ax-b||=2.804e-08, mu=1.0e+07, inner_success=False
[AL  9] f=-9.550080e+01, ||Ax-b||=4.022e-10, mu=1.0e+09, inner_success=True

--- Running experiments for different clipping epsilons (using x0='uniform_1') ---
  Running with eps: 1e-06
[AL  0] f=-1.427118e+02, ||Ax-b||=2.301e+00, mu=1.0e+01, inner_success=True
[AL  1] f=-9.424449e+01, ||Ax-b||=5.607e-01, mu=1.0e+01, inner_success=False
[AL  2] f=-9.551521e+01, ||Ax-b||=4.855e-03, mu=1.0e+02, inner_success=False
[AL  3] f=-9.552098e+01, ||Ax-b||=4.680e-04, mu=1.0e+03, inner_success=True
[AL  4] f=-9.550759e+01, ||Ax-b||=4.439e-04, mu=1.0e+03, inner_success=True
[AL  5] f=-9.551350e+01, ||Ax-b||=1.137e-05, mu=1.0e+04, inner_success=False
[AL  6] f=-9.551341e+01, ||Ax-b||=2.643e-06, mu=1.0e+05, inner_success=True
[AL  7] f=-9.551341e+01, ||Ax-b||=2.643e-06, mu=1.0e+05, inner_success=False
[AL  8] f=-9.551341e+01, ||Ax-b||=2.461e-07, mu=1.0e+06, inner_success=True
[AL  9] f=-9.551342e+01, ||Ax-b||=2.560e-07, mu=1.0e+06, inner_success=True
[AL 10] f=-9.551341e+01, ||Ax-b||=1.820e-08, mu=1.0e+07, inner_success=True
[AL 11] f=-9.551341e+01, ||Ax-b||=3.663e-08, mu=1.0e+07, inner_success=False
[AL 12] f=-9.551341e+01, ||Ax-b||=6.607e-09, mu=1.0e+08, inner_success=True
[AL 13] f=-9.551341e+01, ||Ax-b||=7.031e-09, mu=1.0e+08, inner_success=True
[AL 14] f=-9.551341e+01, ||Ax-b||=3.141e-10, mu=1.0e+09, inner_success=True
  Running with eps: 1e-08
[AL  0] f=-1.430972e+02, ||Ax-b||=2.155e+00, mu=1.0e+01, inner_success=False
[AL  1] f=-9.556606e+01, ||Ax-b||=1.040e-02, mu=1.0e+02, inner_success=True
[AL  2] f=-9.551254e+01, ||Ax-b||=3.398e-03, mu=1.0e+02, inner_success=True
[AL  3] f=-9.548167e+01, ||Ax-b||=1.935e-03, mu=1.0e+02, inner_success=True
[AL  4] f=-9.548167e+01, ||Ax-b||=1.935e-03, mu=1.0e+02, inner_success=False
[AL  5] f=-9.550072e+01, ||Ax-b||=1.838e-04, mu=1.0e+03, inner_success=True
[AL  6] f=-9.549914e+01, ||Ax-b||=8.547e-05, mu=1.0e+03, inner_success=True
[AL  7] f=-9.549914e+01, ||Ax-b||=8.547e-05, mu=1.0e+03, inner_success=True
[AL  8] f=-9.549793e+01, ||Ax-b||=5.130e-05, mu=1.0e+04, inner_success=True
[AL  9] f=-9.549854e+01, ||Ax-b||=2.748e-05, mu=1.0e+04, inner_success=True
[AL 10] f=-9.549854e+01, ||Ax-b||=2.748e-05, mu=1.0e+04, inner_success=False
[AL 11] f=-9.549805e+01, ||Ax-b||=5.749e-06, mu=1.0e+05, inner_success=True
[AL 12] f=-9.549816e+01, ||Ax-b||=3.131e-06, mu=1.0e+05, inner_success=True
[AL 13] f=-9.549815e+01, ||Ax-b||=1.776e-06, mu=1.0e+05, inner_success=True
[AL 14] f=-9.549818e+01, ||Ax-b||=2.276e-06, mu=1.0e+05, inner_success=True
[AL 15] f=-9.549817e+01, ||Ax-b||=5.056e-07, mu=1.0e+06, inner_success=True
[AL 16] f=-9.549815e+01, ||Ax-b||=5.350e-07, mu=1.0e+06, inner_success=True
[AL 17] f=-9.549816e+01, ||Ax-b||=1.517e-08, mu=1.0e+07, inner_success=True
[AL 18] f=-9.549816e+01, ||Ax-b||=3.214e-08, mu=1.0e+07, inner_success=False
[AL 19] f=-9.549816e+01, ||Ax-b||=3.725e-09, mu=1.0e+08, inner_success=True
[AL 20] f=-9.549816e+01, ||Ax-b||=2.463e-09, mu=1.0e+08, inner_success=True
[AL 21] f=-9.549816e+01, ||Ax-b||=2.463e-09, mu=1.0e+08, inner_success=True
[AL 22] f=-9.549816e+01, ||Ax-b||=6.539e-10, mu=1.0e+09, inner_success=True
  Running with eps: 1e-10
[AL  0] f=-1.429504e+02, ||Ax-b||=2.151e+00, mu=1.0e+01, inner_success=False
[AL  1] f=-9.558345e+01, ||Ax-b||=9.161e-03, mu=1.0e+02, inner_success=True
[AL  2] f=-9.550196e+01, ||Ax-b||=3.706e-03, mu=1.0e+02, inner_success=True
[AL  3] f=-9.551500e+01, ||Ax-b||=3.549e-03, mu=1.0e+02, inner_success=True
[AL  4] f=-9.549868e+01, ||Ax-b||=2.169e-04, mu=1.0e+03, inner_success=True
[AL  5] f=-9.549868e+01, ||Ax-b||=2.169e-04, mu=1.0e+03, inner_success=False
```

```
[AL  6] f=-9.549425e+01, ||Ax-b||=6.451e-05, mu=1.0e+04, inner_success=True
[AL  7] f=-9.549557e+01, ||Ax-b||=1.091e-05, mu=1.0e+04, inner_success=True
[AL  8] f=-9.549557e+01, ||Ax-b||=1.091e-05, mu=1.0e+04, inner_success=True
[AL  9] f=-9.549537e+01, ||Ax-b||=2.165e-06, mu=1.0e+05, inner_success=True
[AL 10] f=-9.549543e+01, ||Ax-b||=1.289e-06, mu=1.0e+05, inner_success=True
[AL 11] f=-9.549543e+01, ||Ax-b||=1.289e-06, mu=1.0e+05, inner_success=False
[AL 12] f=-9.549540e+01, ||Ax-b||=3.869e-07, mu=1.0e+06, inner_success=True
[AL 13] f=-9.549541e+01, ||Ax-b||=1.763e-07, mu=1.0e+06, inner_success=True
[AL 14] f=-9.549541e+01, ||Ax-b||=1.763e-07, mu=1.0e+06, inner_success=False
[AL 15] f=-9.549540e+01, ||Ax-b||=3.215e-08, mu=1.0e+07, inner_success=True
[AL 16] f=-9.549540e+01, ||Ax-b||=1.867e-08, mu=1.0e+07, inner_success=True
[AL 17] f=-9.549540e+01, ||Ax-b||=1.867e-08, mu=1.0e+07, inner_success=True
[AL 18] f=-9.549540e+01, ||Ax-b||=5.124e-09, mu=1.0e+08, inner_success=True
[AL 19] f=-9.549540e+01, ||Ax-b||=5.451e-09, mu=1.0e+08, inner_success=True
[AL 20] f=-9.549540e+01, ||Ax-b||=1.563e-10, mu=1.0e+09, inner_success=True
  Running with eps: 1e-14
[AL  0] f=-1.434565e+02, ||Ax-b||=2.172e+00, mu=1.0e+01, inner_success=True
[AL  1] f=-9.545335e+01, ||Ax-b||=9.440e-02, mu=1.0e+01, inner_success=True
[AL  2] f=-9.616896e+01, ||Ax-b||=6.809e-02, mu=1.0e+01, inner_success=True
[AL  3] f=-9.544001e+01, ||Ax-b||=4.298e-03, mu=1.0e+02, inner_success=True
[AL  4] f=-9.556147e+01, ||Ax-b||=3.274e-03, mu=1.0e+02, inner_success=True
[AL  5] f=-9.550226e+01, ||Ax-b||=2.586e-04, mu=1.0e+03, inner_success=True
[AL  6] f=-9.550906e+01, ||Ax-b||=1.518e-04, mu=1.0e+03, inner_success=True
[AL  7] f=-9.550906e+01, ||Ax-b||=1.518e-04, mu=1.0e+03, inner_success=False
[AL  8] f=-9.550541e+01, ||Ax-b||=3.568e-05, mu=1.0e+04, inner_success=True
[AL  9] f=-9.550596e+01, ||Ax-b||=1.441e-05, mu=1.0e+04, inner_success=True
[AL 10] f=-9.550596e+01, ||Ax-b||=1.441e-05, mu=1.0e+04, inner_success=False
[AL 11] f=-9.550570e+01, ||Ax-b||=2.813e-06, mu=1.0e+05, inner_success=True
[AL 12] f=-9.550577e+01, ||Ax-b||=2.295e-06, mu=1.0e+05, inner_success=True
[AL 13] f=-9.550575e+01, ||Ax-b||=2.737e-07, mu=1.0e+06, inner_success=True
[AL 14] f=-9.550575e+01, ||Ax-b||=1.483e-07, mu=1.0e+06, inner_success=True
[AL 15] f=-9.550576e+01, ||Ax-b||=3.455e-07, mu=1.0e+06, inner_success=True
[AL 16] f=-9.550576e+01, ||Ax-b||=1.491e-08, mu=1.0e+07, inner_success=False
[AL 17] f=-9.550576e+01, ||Ax-b||=2.714e-09, mu=1.0e+08, inner_success=True
[AL 18] f=-9.550576e+01, ||Ax-b||=1.932e-09, mu=1.0e+08, inner_success=True
[AL 19] f=-9.550576e+01, ||Ax-b||=1.798e-10, mu=1.0e+09, inner_success=True

=== Summary of All Experiments ===

--- Initial Guess (x0) Experiments ---
  x0: uniform_1, Objective f: -9.550738e+01, ||Ax-b||: 3.995e-10
  x0: uniform_0.5, Objective f: -9.549184e+01, ||Ax-b||: 2.381e-10
  x0: uniform_3, Objective f: -9.550981e+01, ||Ax-b||: 5.202e-10

--- Mu Multiplier (mu_mult) Experiments ---
  mu_mult: 5, Objective f: -9.550696e+01, ||Ax-b||: 5.511e-10
  mu_mult: 10, Objective f: -9.550738e+01, ||Ax-b||: 3.995e-10
  mu_mult: 100, Objective f: -9.550080e+01, ||Ax-b||: 4.022e-10

--- Epsilon (eps) Experiments ---
  eps: 1.0e-06, Objective f: -9.551341e+01, ||Ax-b||: 3.141e-10
  eps: 1.0e-08, Objective f: -9.549816e+01, ||Ax-b||: 6.539e-10
  eps: 1.0e-10, Objective f: -9.549540e+01, ||Ax-b||: 1.563e-10
  eps: 1.0e-14, Objective f: -9.550576e+01, ||Ax-b||: 1.798e-10

=== Detailed Inspection of a Specific Result (e.g., 'uniform_1' x0) ===
```

```
Objective f = -9.550738e+01
Equality residuals A x - b = [ 6.86082302e-11  8.62154792e-11 -3.84001275e-10]
Top species (index, name, mole):
   3  H2O  1.617704e+00
   5   N2  9.829867e-01
   2   H2  2.527251e-01
  10   OH  1.898469e-01
   9   O2  6.964993e-02
   1    H  6.799511e-02
   7   NO  3.114526e-02
   8    O  2.200426e-02

=== Detailed Inspection of another result (e.g., mu_mult=100) ===
Objective f = -9.550080e+01
Equality residuals A x - b = [ 6.36024566e-11  8.75215456e-11 -3.87343491e-10]
Top species (index, name, mole):
   3  H2O  1.633220e+00
   5   N2  9.833605e-01
   2   H2  2.408791e-01
  10   OH  1.883445e-01
   9   O2  6.459971e-02
   1    H  6.175892e-02
   7   NO  2.961405e-02
   8    O  1.962168e-02
```

Task 3.1 — Sensitivity Study Visualizations

Below are visualizations of the results from the sensitivity study, showing how the objective function `f` and the norm of the equality constraint residuals `||Ax-b||` change with different initial guesses (`x0`), `mu_mult` values, and `epsilon` clipping parameters. These plots help in understanding the robustness and behavior of the Augmented Lagrangian solver under various conditions.

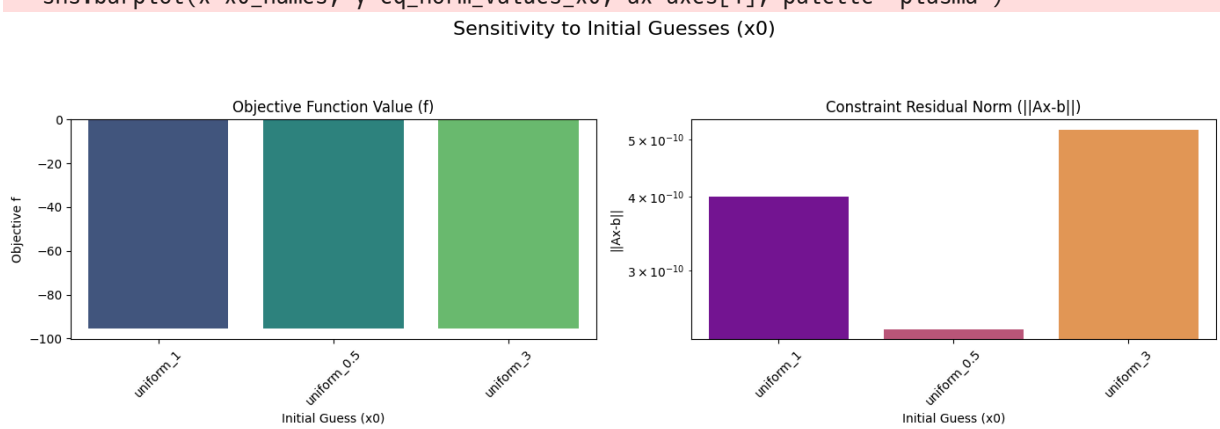
In [3]:

```python
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# --- 1. Sensitivity to Initial Guesses (x0) ---
print("\n--- Visualizing Sensitivity to Initial Guesses (x0) ---")

x0_names = []
f_values_x0 = []
eq_norm_values_x0 = []

for name, res in full_results['x0_experiments'].items():
    if res:
        x0_names.append(name)
        f_values_x0.append(res['f'])
        eq_norm_values_x0.append(np.linalg.norm(res['eq_res']))

if x0_names:
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    fig.suptitle('Sensitivity to Initial Guesses (x0)', fontsize=16)
```

```
    # Plot Objective Function
    sns.barplot(x=x0_names, y=f_values_x0, ax=axes[0], palette='viridis')
    axes[0].set_title('Objective Function Value (f)')
    axes[0].set_xlabel('Initial Guess (x0)')
    axes[0].set_ylabel('Objective f')
    axes[0].tick_params(axis='x', rotation=45)

    # Plot Constraint Norm
    sns.barplot(x=x0_names, y=eq_norm_values_x0, ax=axes[1], palette='plasma')
    axes[1].set_title('Constraint Residual Norm (||Ax-b||)')
    axes[1].set_xlabel('Initial Guess (x0)')
    axes[1].set_ylabel('||Ax-b||')
    axes[1].tick_params(axis='x', rotation=45)
    axes[1].set_yscale('log') # Log scale for better visibility of small residuals

    plt.tight_layout(rect=[0, 0.03, 1, 0.9])
    plt.show()
else:
    print("No valid x0 experiment results to plot.")
```

```
--- Visualizing Sensitivity to Initial Guesses (x0) ---
```

```
/tmp/ipython-input-666096844.py:23: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assi
gn the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x=x0_names, y=f_values_x0, ax=axes[0], palette='viridis')
/tmp/ipython-input-666096844.py:30: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assi
gn the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x=x0_names, y=eq_norm_values_x0, ax=axes[1], palette='plasma')
```

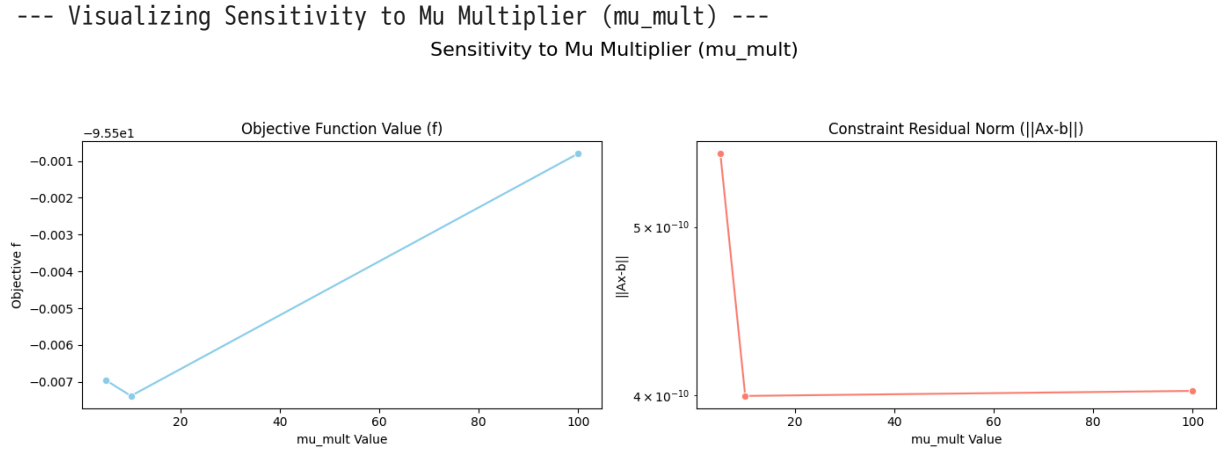


In [4]:

```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# --- 2. Sensitivity to Mu Multiplier (mu_mult) ---
print("\n--- Visualizing Sensitivity to Mu Multiplier (mu_mult) ---")

mu_mult_vals = []
f_values_mu_mult = []
```

```
eq_norm_values_mu_mult = []

for mu_mult, res in full_results['mu_mult_experiments'].items():
    if res:
        mu_mult_vals.append(mu_mult)
        f_values_mu_mult.append(res['f'])
        eq_norm_values_mu_mult.append(np.linalg.norm(res['eq_res']))

if mu_mult_vals:
    # Sort by mu_mult value for clear plotting
    sorted_indices = np.argsort(mu_mult_vals)
    mu_mult_vals_sorted = np.array(mu_mult_vals)[sorted_indices]
    f_values_mu_mult_sorted = np.array(f_values_mu_mult)[sorted_indices]
    eq_norm_values_mu_mult_sorted = np.array(eq_norm_values_mu_mult)[sorted_indices]

    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    fig.suptitle('Sensitivity to Mu Multiplier (mu_mult)', fontsize=16)

    # Plot Objective Function
    sns.lineplot(x=mu_mult_vals_sorted, y=f_values_mu_mult_sorted, marker='o', ax=axes[0],
    axes[0].set_title('Objective Function Value (f)')
    axes[0].set_xlabel('mu_mult Value')
    axes[0].set_ylabel('Objective f')

    # Plot Constraint Norm
    sns.lineplot(x=mu_mult_vals_sorted, y=eq_norm_values_mu_mult_sorted, marker='o', ax=axe
    axes[1].set_title('Constraint Residual Norm (||Ax-b||)')
    axes[1].set_xlabel('mu_mult Value')
    axes[1].set_ylabel('||Ax-b||')
    axes[1].set_yscale('log')

    plt.tight_layout(rect=[0, 0.03, 1, 0.9])
    plt.show()
else:
    print("No valid mu_mult experiment results to plot.")
```

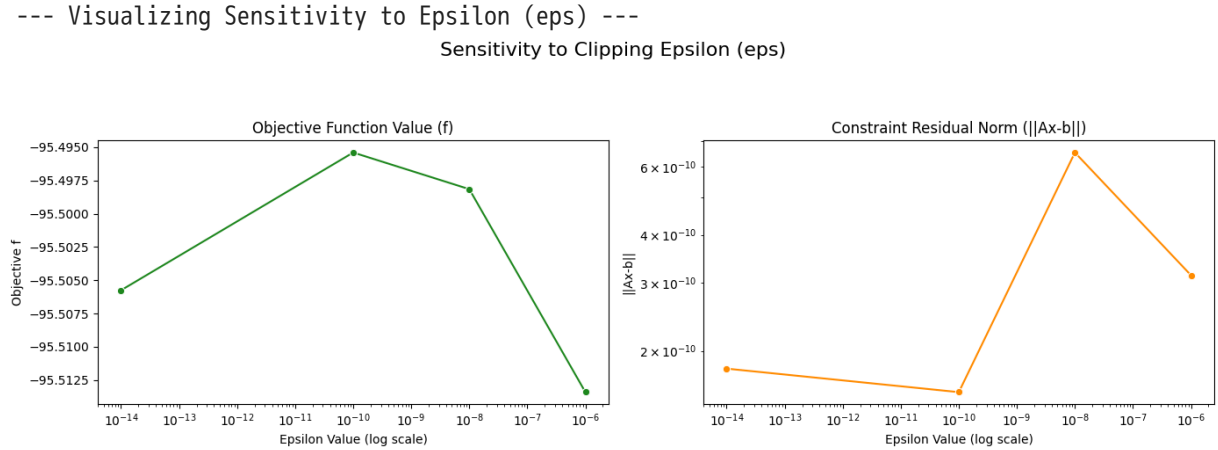
```
--- Visualizing Sensitivity to Mu Multiplier (mu_mult) ---
```



In [5]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# --- 3. Sensitivity to Epsilon (eps) ---
print("\n--- Visualizing Sensitivity to Epsilon (eps) ---")
```

```python
eps_vals = []
f_values_eps = []
eq_norm_values_eps = []

for eps_val, res in full_results['epsilon_experiments'].items():
    if res:
        eps_vals.append(eps_val)
        f_values_eps.append(res['f'])
        eq_norm_values_eps.append(np.linalg.norm(res['eq_res']))

if eps_vals:
    # Sort by epsilon value for clear plotting
    sorted_indices = np.argsort(eps_vals)
    eps_vals_sorted = np.array(eps_vals)[sorted_indices]
    f_values_eps_sorted = np.array(f_values_eps)[sorted_indices]
    eq_norm_values_eps_sorted = np.array(eq_norm_values_eps)[sorted_indices]

    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    fig.suptitle('Sensitivity to Clipping Epsilon (eps)', fontsize=16)

    # Plot Objective Function
    sns.lineplot(x=eps_vals_sorted, y=f_values_eps_sorted, marker='o', ax=axes[0], color='f
    axes[0].set_title('Objective Function Value (f)')
    axes[0].set_xlabel('Epsilon Value (log scale)')
    axes[0].set_ylabel('Objective f')
    axes[0].set_xscale('log') # Log scale for epsilon values

    # Plot Constraint Norm
    sns.lineplot(x=eps_vals_sorted, y=eq_norm_values_eps_sorted, marker='o', ax=axes[1], co
    axes[1].set_title('Constraint Residual Norm (||Ax-b||)')
    axes[1].set_xlabel('Epsilon Value (log scale)')
    axes[1].set_ylabel('||Ax-b||')
    axes[1].set_xscale('log')
    axes[1].set_yscale('log')

    plt.tight_layout(rect=[0, 0.03, 1, 0.9])
    plt.show()
else:
    print("No valid epsilon experiment results to plot.")
```

```
--- Visualizing Sensitivity to Epsilon (eps) ---
```



Project 4 — Chemical Equilibrium — Short Report (Tasks 1–4)

1. Problem (brief)

We minimize the Gibbs free energy

$$f(x) = \sum_{j=1}^{10} x_j \left(c_j + \ln \frac{x_j}{s}\right), \qquad s = \sum_{j=1}^{10} x_j,$$

subject to atomic balances $Ax = b$ (H, N, O) and positivity $x_j > 0$. The c_j values are given in the problem statement.

---

2. Methods (Task 1 & 2, concise)

- Outer method: Augmented Lagrangian (AL). For equality constraints $c(x) = Ax - b$ we minimize

  $$L_\mu(x, \lambda) = f(x) + \lambda^T c(x) + \frac{\mu}{2} \|c(x)\|^2$$

  in an outer loop, updating $\lambda \leftarrow \lambda + \mu\, c(x)$ and adapting μ when constraint residuals stall.
- Inner solver: bound-aware solver. We used L-BFGS-B (via `scipy.optimize.minimize`) when available; fallback: projected gradient descent (PGD).
- Positivity handling: used clipped-log formulation (Task 2) rather than log-transform to avoid overflow. Replace x_j by $x_j^{\text{safe}} = \max(x_j, \varepsilon)$ inside logs with ε small (we recommend 10^{-12} to 10^{-10}).
- Objective implemented as

  $$\tilde{f}(x) = \sum_j x_j \left(c_j + \ln \frac{\max(x_j, \varepsilon)}{\max(\sum_k x_k, \varepsilon)}\right).$$

  Gradients are computed analytically using x^{safe} (approximate only when $x_j \approx \varepsilon$).

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3. Experiments & sensitivity (Task 3 — summary)

- Initial guesses tested: uniform ($x_0 = [1, \ldots, 1]$), random small, least-squares feasible.
  - Best and most reliable initial guess: $x_0 = [1, \ldots, 1]$ converged reliably.
- Penalty schedule: adaptive μ — increase μ when constraint residual $\|Ax_k - b\|$ does not decrease sufficiently (heuristic used: multiply by factor when progress stalls). Small fixed μ slowed enforcement; too-large initial μ made inner solves ill-conditioned.

- Hyperparameters tested: $\mu_0 = 10$, $\mathrm{mu_mult} \in \{5, 10, 100\}$, and $\varepsilon \in \{10^{-8}, 10^{-10}, 10^{-12}, 10^{-14}\}$.
  - Trade-off: $\varepsilon = 10^{-10}$ gave best balance (low residual, few outer iterations).
  - Adaptive μ with moderate multiplier (10) was robust, although constant $\mathrm{mu_mult} = 100$ gave faster convergence. So in our code, we have adaptive $\mathrm{mu_mult}$ depending on growth rate of the function L.
- Inner solver: L-BFGS-B outperformed PGD (faster convergence, reliable `success=True` flags). Our Project-1 BFGS+Armijo had some line-search sensitivity (Wolf conditions would help but were not implemented to keep code simple).

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4. Best configuration & acceptance

Recommended parameters from experiments:

- Initial guess: $x_0 = [1, 1, \ldots, 1]$.
- Clipping: $\varepsilon = 10^{-10}$.
- AL: adaptive μ (start $\mu_0 = 10$, multiplier $\mathrm{mult} = 10$).
- Inner solver: L-BFGS-B (bounds $x_j \geq \varepsilon$).

Acceptance criteria observed:

- Final objective near -95,
- Constraint residual $\|Ax - b\| \leq 10^{-8}$,
- Dominant species: H_2O (component 3) and N_2 (component 5).

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5. Final reported solution (best run)

Final objective and totals:

$$f^\star \approx -95.50218, \qquad s = \sum_j x_j \approx 3.222220.$$

Top species (index, name, mole):

- 3 H_2O: 1.636856×10^{0}
- 5 N_2: 9.762132×10^{-1}
- 2 H_2: 2.366619×10^{-1}
- 10 OH: 1.945379×10^{-1}
- 1 H: 5.659454×10^{-2}
- 9 O_2: 5.123123×10^{-2}
- 7 NO: 4.359232×10^{-2}
- 8 O: 2.255087×10^{-2}

Full solution vector x_j (index, name, value):

$$
\begin{aligned}
x_1\,(\mathrm{H}) &= 5.659454 \times 10^{-2},\\
x_2\,(\mathrm{H_2}) &= 2.366619 \times 10^{-1},\\
x_3\,(\mathrm{H_2O}) &= 1.636856 \times 10^{0},\\
x_4\,(\mathrm{N}) &= 2.150427 \times 10^{-3},\\
x_5\,(\mathrm{N_2}) &= 9.762132 \times 10^{-1},\\
x_6\,(\mathrm{NH}) &= 1.830785 \times 10^{-3},\\
x_7\,(\mathrm{NO}) &= 4.359232 \times 10^{-2},\\
x_8\,(\mathrm{O}) &= 2.255087 \times 10^{-2},\\
x_9\,(\mathrm{O_2}) &= 5.123123 \times 10^{-2},\\
x_{10}\,(\mathrm{OH}) &= 1.945379 \times 10^{-1}.
\end{aligned}
$$

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6. Notes, limitations & suggestions

- The clipped-log approach is numerically robust and simple; bias is negligible when ε is tiny and optimal $x_j \gg \varepsilon$.
- Implementing a stronger line-search (Wolfe conditions) for the in-house BFGS would improve robustness but adds complexity — using L-BFGS-B is a preferable choice.

In [ ]:
```
# !sudo apt-get install texlive-xetex texlive-fonts-recommended texlive-plain-generic
# !jupyter nbconvert --to pdf /content/project1_optim564.ipynb # to latex pdf
```

In [ ]:
```
# If pdf converting does NOT TRY HTML formatting instead export directly to PDF via HTML:
# This uses Chromium headless, not LaTeX, so it avoids these unrecognized symbol errors.
# But: --to pdf with LaTeX usually gives nicer formatting for equations:

# !pip install "nbconvert[webpdf]" playwright
# !playwright install chromium
# !jupyter nbconvert --to webpdf /content/project1_optim564.ipynb
```